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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 5

Value-added-services

Objectives

By the end of this topic you should be able to;

- *Understand Mobile communications*
- *Give the various existing mobile communication technologies*
- *Understand value added services*

5.1. Mobile Communications

Since the time of wireless telegraphy, radio communication has been used extensively. Initially the mobile communication was limited between one pair of users on single channel pair. The range of mobility was defined by the transmitter power, type of antenna used and the frequency of operation. With the increase in the number of users, accommodating them within the limited available frequency spectrum became a major problem. To resolve this problem, the concept of cellular communication was evolved. The present day cellular communication uses a basic unit called cell. Areas are divided into cells and each is served by its own antenna. It is also served by base station consisting of transmitter, receiver, and control unit. Cells are allocated bands of frequencies. The cells are set up such that antennas of all neighbors are equidistant (hexagonal pattern). Each cell consists of small hexagonal area with a base station located at the center of the cell which communicates with the user. To accommodate multiple users Time Division multiple Access (TDMA), Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA) and their hybrids are used. Numerous mobile radio standards have been deployed at various places such as AMPS & PACS.

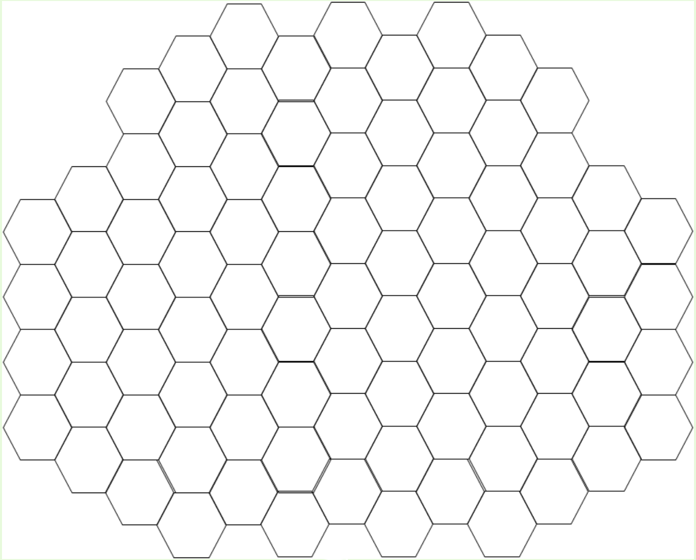
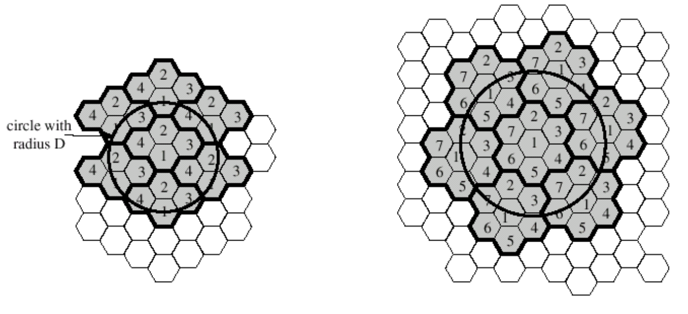


Fig 5.1: Hexagonal Cells

5.1.1. Frequency Reuse

Frequency reuse, or, frequency planning, is a technique of reusing frequencies and channels within a communication system to improve capacity and spectral efficiency. Frequency reuse is one of the fundamental concepts on which commercial wireless systems are based that involve the partitioning of an RF radiating area into cells. The increased capacity in a commercial wireless network, compared with a network with a single transmitter, comes from the fact that the same radio frequency can be reused in a different area for a completely different transmission. Frequency reuse in mobile cellular systems means that frequencies allocated to adjacent cells are assigned different frequencies to avoid interference or crosstalk. The main objective is to reuse frequency in nearby cells 10 to 50 frequencies assigned to each cell. Transmission power is controlled to limit power at that frequency escaping to adjacent cells. It determines how many cells must intervene between two cells using the same frequency. In a frequency reuse, the total number of channels are divided into K groups. K is called reuse factor or cluster size. Each cell is assigned one of the groups. The same group can be reused by two different cells provided that they are sufficiently far apart.



(a) Frequency reuse pattern for $N = 4$ (b) Frequency reuse pattern for $N = 7$

The repeating regular pattern of cells is called cluster. Since each cell is designed to use radio frequencies only within its boundaries, the same frequencies can be reused in other cells not far away without interference, in another cluster. Such cells are called 'co-channel' cells. The reuse of frequencies enables a cellular system to handle a huge number of calls with a limited

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number of channels. The figure above shows a frequency planning with cluster size of 4 and 7, showing the co-channels cells in different clusters by the same letter. The closest distance between the co-channel cells (in different clusters) is determined by the choice of the cluster size and the layout of the cell cluster. Consider a cellular system with S duplex channels available for use and let N be the number of cells in a cluster. If each cell is allotted K duplex channels with all being allotted unique and disjoint channel groups we have S = KN under normal circumstances. Now, if the cluster are repeated M times within the total area, the total number of duplex channels, or, the total number of users in the system would be T = MS = KMN.

Example ✎. Assume a network system with 32 cells of radius 1.6 km and a total frequency bandwidth that supports 336 traffic channels and reuse factor n=7. What geographical area is covered and how many channels are there per cell and what is the number of occurent calls that can be handled?

Solution:

$Area\ of\ hexagonal\ = 1.5R\sqrt[2]{3}$

$Radius\ = 1.6\ km$

$1.5(1.6)^2\sqrt{3} = 6.65km^2$

$= 32 * 6.65 = 213km^2(areacoveredbythenetwork)$

$336channels:$

$reusable\ factor = 7$

$\frac{336}{7} = 48\ channels$

$Total\ Capacity\ = 48 * 32 = 1536\ calls$

□

5.1.2. How to cope with increasing capacity of cells

- *Cell splitting:* Cell Splitting is based on the cell radius reduction and minimizes the need to modify the existing cell parameters. Cell splitting involves the process of sub-dividing a congested cell into smaller cells, each with its own base station and a corresponding reduction in antenna size and transmitting power. This increases the capacity of a cellular system since it increases the number of times that channels are reused.
- *Cell sectoring:* Sectoring is basically a technique which can increase the SIR without necessitating an increase in the cluster size.
- By adding new channels

- Frequency borrowing from adjacent cells
- Microcells: Antennas can be moved to buildings, hills, and lamp posts

5.2. Mobile Radio Transmission Techniques

1. **Simplex system**

   Simplex systems utilize simplex channels i.e., the communication is unidirectional. The first user can communicate with the second user. However, the second user cannot communicate with the first user. One example of such a system is a pager.

2. **Half duplex system**

   Half duplex radio systems that use half duplex radio channels allow for non-simultaneous bidirectional communication. The first user can communicate with the second user but the second user can communicate to the first user only after the first user has finished his conversation. At a time, the user can only transmit or receive information. A walkie-talkie is an example of a half duplex system which uses 'push to talk' and 'release to listen' type of switches.

3. **Full duplex system**

   Full duplex systems allow two way simultaneous communications. Both the users can communicate to each other simultaneously. This can be done by providing two simultaneous but separate channels to both the users. This is possible by one of the two following methods.

- **Frequency Division Duplexing (FDD):** FDD supports two-way radio communication by using two distinct radio channels. One frequency channel is transmitted downstream from the BS to the MS (forward channel). A second frequency is used in the upstream direction and supports transmission from the MS to the BS (reverse channel). Because of the pairing of frequencies, simultaneous transmission in both directions is possible. To mitigate self-interference between upstream and downstream transmissions, a minimum amount of frequency separation must be maintained between the frequency pair, telephone line with a specific telephone number on the Public Switched Telephone Network (PSTN). A mobile system, in general, on the other hand, is the example of the second category of a full duplex mobile system where many users connect among themselves via a single BS.

- **Time Division Duplexing (TDD):** TDD uses a single frequency band to transmit signals in both the downstream and upstream directions. TDD operates by toggling transmission directions over a time interval. This toggling takes place very rapidly and is imperceptible to the user. A full duplex mobile system can further be subdivided into two category: a single MS for a dedicated BS, and many MS for a single BS. Cordless telephone systems are full duplex communication systems that use radio to connect to a portable handset to a single dedicated BS, which is then connected to a dedicated telephone line with a specific telephone number on the Public Switched Telephone Network (PSTN). A mobile system, in general, on the other hand, is the example of the second category of a full duplex mobile system where many users connect among themselves via a single BS.

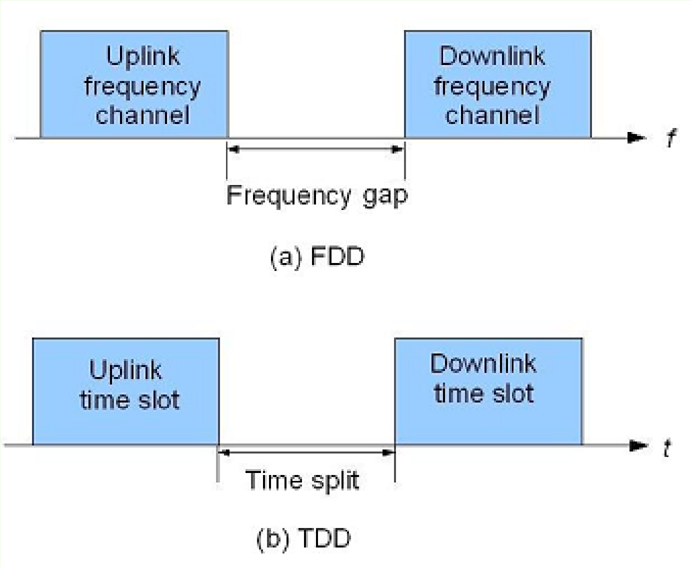

Figure 5.3: (a) Frequency division duplexing and (b) time division duplexing.

5.2.1. How a Mobile Call is Made

In order to know how a mobile call is made, we should first look into the basics of cellular concept and main operational channels involved in making a call. These are given below.

- **Cellular Concept**

Cellular telephone systems must accommodate a large number of users over a large geographic area with limited frequency spectrum, i.e., with limited number of channels. If a single transmitter/ receiver is used with only a single base station, then sufficient amount of power may not be present at a huge distance from the BS.

For a large geographic coverage area, a high powered transmitter therefore has to be used. But a high power radio transmitter causes harm to environment. Mobile communication thus calls for replacing the high power transmitters by low power transmitters by dividing the coverage area into small segments, called cells. Each cell uses a certain number of the available channels and a group of adjacent cells together use all the available channels. Such a group is called a cluster. This cluster can repeat itself and hence the same set of channels can be used again and again.

Each cell has a low power transmitter with a coverage area equal to the area of the cell. This technique of substituting a single high powered transmitter by several low powered transmitters to support many users is the backbone of the cellular concept.

- **Operational Channels**

In each cell, there are four types of channels that take active part during a mobile call. These are:

- **Forward Voice Channel (FVC):** This channel is used for the voice transmission from the BS to the MS.
- **Reverse Voice Channel (RVC):** This is used for the voice transmission from the MS to the BS.
- **Forward Control Channel (FCC):** Control channels are generally used for controlling the activity of the call, i.e., they are used for setting up calls and to divert the call to unused voice channels. Hence these are also called setup channels. These channels transmit and receive call initiation and service request messages. The FCC is used

for control signaling purpose from the BS to MS.

- **Reverse Control Channel (RCC):** This is used for the call control purpose from the MS to the BS. Control channels are usually monitored by mobiles.

- **Making a Call**

When a mobile is idle, i.e., it is not experiencing the process of a call, then it searches all the FCCs to determine the one with the highest signal strength. The mobile then monitors this particular FCC. However, when the signal strength falls below a particular threshold that is insufficient for a call to take place, the mobile again searches all the FCCs for the one with the highest signal strength. For a particular country or continent, the control channels will be the same. So all mobiles in that country or continent will search among the same set of control channels. However, when a mobile moves to a different country or continent, then the control channels for that particular location will be different and hence the mobile will not work.

Each mobile has a mobile identification number (MIN). When a user wants to make a call, he sends a call request to the MSC on the reverse control channel. He also sends the MIN of the person to whom the call has to be made. The MSC then sends this MIN to all the base stations. The base station transmits this MIN and all the mobiles within the coverage area of that base station receive the MIN and match it with their own. If the MIN matches with a particular MS, that mobile sends an acknowledgment to the BS. The BS then informs the MSC that the mobile is within its coverage area. The MSC then instructs the base station to access specific unused voice channel pair. The base station then sends a message to the mobile to move to the particular channels and it also sends a signal to the mobile for ringing.

In order to maintain the quality of the call, the MSC adjusts the transmitted power of the mobile which is usually expressed in dB or dBm. When a mobile moves from the coverage area of one base station to the coverage area of another base station i.e., from one cell to another cell, then the signal strength of the initial base station may not be sufficient to continue the call in progress. So the call has to be transferred to the other base station. This is called handoff. In such cases, in order to maintain the call, the MSC transfers the call to one of the unused voice channels of the new base station or it transfers the control of the current voice channels to the new base station.

5.3. Mobile propagation effects

1. ***Signal strength***: The mobile signal must be strong enough between base station and mobile unit to maintain signal quality at the receiver. The strength must be so strong so as to create co-channel interference with channels in another cell using the same frequency band.
2. ***Fading:*** This is where the signal propagation effects may disrupt the signal and cause errors.
3. ***Handoff:*** A handoff is the term used to describe migration from one BS to another in a mobile network. CDMA uses soft handoff while GSM use hard handoff.

- Soft Handoff: commences communication with a new BS without interrupting communication with old BS. It uses the same frequency assignment between old and new BS provides different site selection diversity.
- Hard handoff: this is the transmission between two base stations with different frequency assignment.

5.3.1. Handoff Performance Metrics

- Call blocking probability: probability of a new call being blocked
- Call dropping probability: probability that a call is terminated due to a handoff
- Call completion probability: probability that an admitted call is not dropped before it terminates

- Probability of unsuccessful handoff: probability that a handoff is executed while the reception conditions are inadequate
- Handoff blocking probability: probability that a handoff cannot be successfully completed
- Handoff probability: probability that a handoff occurs before call termination
- Rate of handoff: number of handoffs per unit time
- Interruption duration: duration of time during a handoff in which a mobile is not connected to either base station
- Handoff delay: distance the mobile moves from the point at which the handoff should occur to the point at which it does occur.

5.3.2. Factors Influencing Handoffs

The following factors influence the entire handoff process:

1. **Transmitted power:** The transmission power is different for different cells, the handoff threshold or the power margin varies from cell to cell.
2. **Received power:** the received power mostly depends on the Line of Sight (LoS) - path between the user and the BS. Especially when the user is on the boundary of the two cells, the LoS path plays a critical role in handoffs and therefore the power margin _ depends on the minimum received power value from cell to cell.
3. **Area and shape of the cell:** Apart from the power levels, the cell structure also a plays an important role in

the handoff process.

4. **Mobility of users:** The number of mobile users entering or going out of a particular cell, also fixes the handoff strategy of a cell.

5.3.3. Handoffs In Different Generations

In 1G analog cellular systems, the signal strength measurements were made by the BS and in turn supervised by the MSC. The handoffs in this generation can be termed as Network Controlled Hand-Off (NCHO). The BS monitors the signal strengths of voice channels to determine the relative positions of the subscriber.

The special receivers located on the BS are controlled by the MSC to monitor the signal strengths of the users in the

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neighbouring cells which appear to be in need of handoff. Based on the information received from the special receivers the MSC decides whether a handoff is required or not. The approximate time needed to make a handoff successful was about 5-10 s. This requires the value to be in the order of 6dB to 12dB.

In the 2G systems, the MSC was relieved from the entire operation. In this generation, which started using the digital technology, handoff decisions were mobile assisted and therefore called Mobile Assisted Hand-Off (MAHO). In MAHO, the mobile center measures the power changes received from nearby base stations and notifies the two BS. Accordingly the two BS communicate and channel transfer occurs. As compared to 1G, the circuit complexity was increased here whereas the delay in handoff was reduced to 1-5 s. The value was in the order of 0-5

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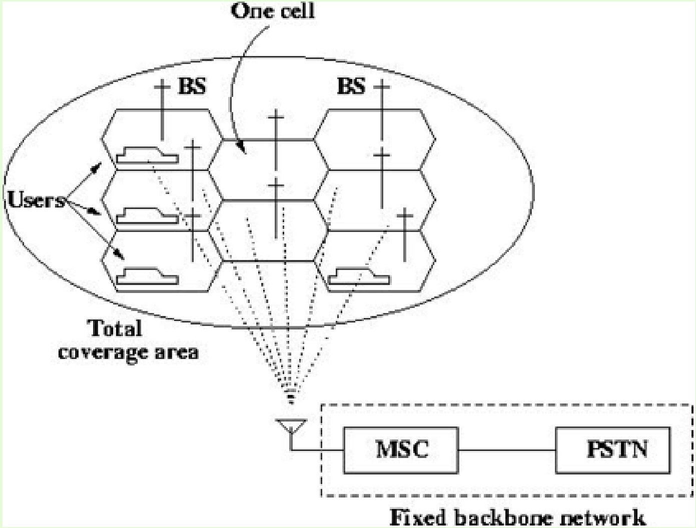
dB.

However, even this amount of delay could create a communication pause. In the current 3G systems, the MS measures the power from adjacent BS and automatically upgrades the channels to its nearer BS. Hence this can be termed as Mobile Controlled Hand-Off (MCHO). When compared to the other generations, delay during handoff is only 100 ms and the value is around 20 dBm. The Quality Of Service (QoS) has improved a lot although the complexity of the circuitry has further increased which is inevitable.

All these types of handoffs are usually termed as hard handoff as there is a shift in the channels involved. There is also another kind of handoff, called soft handoff as discussed below.

Handoff in CDMA: In spread spectrum cellular systems, the mobiles share the same channels in every cell. The MSC evaluates the signal strengths received from different BS for a single user and then shifts the user from one BS to the other without actually changing the channel. These types of handoffs are called soft handoff as there is no change in the channel.


One cell
BS
BS
Users
Total
coverage area
MSC
PSTN
Fixed backbone network

5.3.4. Mobile communication techniques

By definition, mobile radio terminal means any radio terminal that could be moved during its operation. Depending on the radio channel, there can be three different types of mobile communication. In general, however, a Mobile Station (MS) or subscriber unit communicates to a fixed Base Station (BS) which in turn communicates to the desired user at the other end. The MS consists of transceiver, control circuitry, duplexer and an antenna while the BS consists of transceiver and channel multiplexer along with antennas mounted on the tower. The BS are also linked to a power source for the transmission of the radio signals for communication and are connected to a fixed backbone network. The figure above shows a basic mobile communication with low power transmitters/receivers at the BS, the MS and also the Mobile Switching Center (MSC). The MSC is sometimes also called Mobile Telephone Switching Office (MTSO). The radio signals emitted by the BS decay as the signals travel away from it. A minimum amount of signal strength is needed in order to be detected by the mobile stations or mobile sets which are the hand-held personal units (portables) or those installed in the vehicles (mobiles). The region over which the signal strength lies above such a threshold value is known as the coverage area of a BS. The fixed backbone network is a wired network that links all the base stations and also the landline and other telephone networks through wires.

5.4. Existing Mobile Communication Technologies

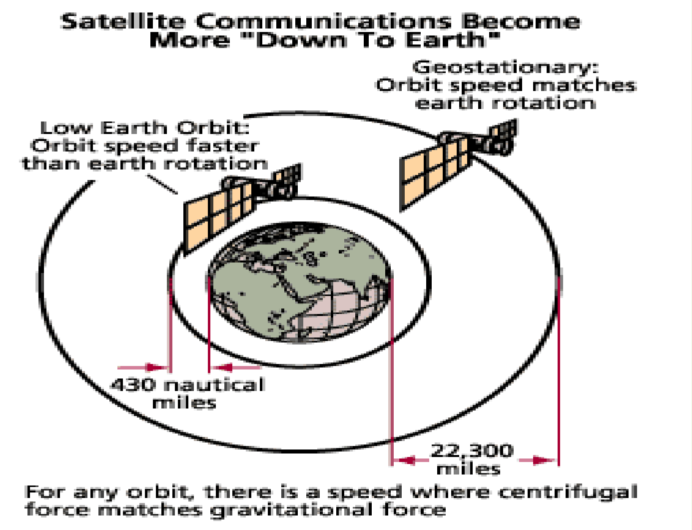
The cellular radio network system facilitates mobility in communication. Systems achieve mobility by transmitting data via radio waves. The following are some examples of mobile communication systems currently in use.

5.4.1. Paging (SMS)

A simple and inexpensive form of mobile communication. An antenna or satellite broadcasts short messages to subscribers. Receivers are usually devices such as beepers, which display messages on a small screen. Transmission of data is one-way. Paging systems are designed to provide reliable communication to subscribers wherever they are. This necessitates high-powered transmitters and low data rates for maximum coverage of each transmitter's designated area.

5.4.2. Communication Satellites

Satellites consist of large transponders that listen to a particular radio frequency, amplify the signal, and then rebroadcast it at another frequency. They are inherently broadcast devices. A drawback of satellites is that they have quite a large propagation delay due to the distances traveled by radio waves.


Satellite Communications Become More "Down To Earth"
Geostationary: Orbit speed matches earth rotation
Low Earth Orbit: Orbit speed faster than earth rotation
430 nautical miles
22,300 miles
For any orbit, there is a speed where centrifugal force matches gravitational force

5.4.3. Cellular Radio Networks

Cellular networks are divided up into cells, each cell being serviced by one or more radio transceivers (transmitter/receiver). Communication in a cellular network is full duplex, where communication is attained by sending and receiving messages on two different frequencies - frequency division duplexing (FDD). The reason for the cellular topology of the network is to enable frequency reuse. Cells, a certain distance apart, can reuse the same frequencies, which ensure the efficient usage of limited radio resources.

5.4.4. Personal Handyphone

The Personal Handyphone System (PHS) is used in Japan. It is similar to cellular networks, however phones can also communicate directly with one another when in range. This is an advantage over cellular phones, which can only communicate with one another via base station transceivers. This system is very popular within heavily populated metropolitan areas.

5.4.5. Mobile Radio

Mobile radio is in many ways the predecessor to the cellular radio network. It is mostly analogue, and makes use of single frequencies for sending and receiving signals. Communication is half-duplex, and a button must be pressed to switch modes. They are most commonly used for emergency services, the transport sector, and the security industry.

5.5. Value Added Services

Value added services is a term that is used to refer to service options that are complimentary to but also ancillary to a core service offering. Value added services are often introduced to customers after the client has purchased the core service that these ancillary offerings center around.

In some instances, a value added service is something extra that is provided to a customer at no additional charge. At other times, the ancillary service is offered to an existing customer as an added service that is available for a modest additional fee. The actual pricing structure for value added services will usually depend on whether the providers sees the services as amenities that are intended to create a stronger rapport with customers or as an additional revenue stream.

Value added services provide advantages for both the customer and the service provider. Customers have the opportunity to receive something above and beyond their basic needs. Providers benefit from the increased rapport with the client that is likely to translate into a more consistent flow of revenue. These additional custom services often cost the provider little to nothing to provide, yet have the potential to enhance the growth and the reputation of the company significantly.

Cellular network operators have to be at the forefront of technology to survive the extremely competitive cellular network environment. Network infrastructures must continually be upgraded and new applications and protocols need to be developed that provide novel services to customers.

All VAS share the same characteristics:

- Not a form of basic service but rather adds value total service offering
- Stands alone in terms of profitability and/or stimulates incremental demand for core service(s)
- Can sometimes stand alone operationally
- Does not affect basic service unless clearly favorable
- Can be an add-on to basic service, and as such, may be sold at a premium price
- May provide operational and/or administrative synergy between or among other services – not merely for diversification.

5.5.1. WAP

WAP enables wireless devices to transfer information in a reliable and effective manner. The constraints of wireless medium forced developers to create a protocol that allows for the low bandwidth and unreliability of the wireless medium to be used effectively. WAP has an optimized protocol stack, and uses a mark-up language at the application layer - WML and WMLScript. It uses UDP instead of TCP, due to the unreliability of the radio link. It has transaction and security layers, which were included to enable the development of applications allowing secure business transactions.

The WAP model interposes a WAP gateway between traditional web servers and WAP clients. The gateway translates HTML into WML, and compresses it into a binary form that is used to save radio-link bandwidth. The client has a WAP browser, which interprets the compressed binary files.

5.5.2. Push-to-Talk

Push-to-Talk (PTT) is a VAS because it:

- Drives additional revenue to the wireless carrier, but doe not cannibalize existing revenues
- Provides differentiated service offerings
- May be packaged with various other VAS such as MIM to provide even greater value

5.5.3. Call Management Services

This type of service can not stand alone as a service. Instead, it adds value to a core service by allowing the subscriber to manage incoming and/or outgoing calls. For example, value-added service interactions occur when the subscriber receives a call. Many call management services allow the subscriber to establish when, where, and under what circumstances they may be reached by calling parties. This provides value to the core service - voice communications - by way of increased control and flexibility.

Depending on the specific commercial situation, this value-added service could be offered as either a premium service (at a premium price) or be bundled with other the core service offering. The benefit of bundling would be to provide a differentiated core service and/or to increase the use of the core service.

5.5.4. Location Sensitive Billing

This is another example of a service that can not stand-alone. Instead, location sensitive billing (LSB) adds value to the core service by location enabling the core service. Location sensitive billing can be used in conjunction with post-paid, prepaid, and/or VPN based mobile communications services to establish zones for which differentiated billing treatment may be applied. For example, a "home zone", "work zone", and "premium price zone" could be established to allow an operator to offer differentiated service to its customers.

This is viewed as a value-added service to both the customer and the mobile operator. The customer benefits from LSB through his ability to use the mobile phone at preferred rates based on location. The wireless carrier benefits from incremental revenues derived from additional usage and from premium charge zones where there is already high demand and perhaps overly taxed system capacity. While the issue of potential cannibalization of existing service arises, customer behavior and studies indicate a net benefit derived from overall increased usage and revenues.

Taken together, call management services and LSB also depict characteristic number six, operational synergy. Call management services add value in terms of providing the user options depending on location. For example, the user may want to receive certain calls at the home zone, but not at work, and perhaps receive only urgent calls when traveling or on vacation. LSB provides the additional synergistic benefit of location based billing when the user is in those various locations.

5.5.5. Mobile Data Services

This is an example of a value-added service that does stand-alone. Mobile data services are considered value-added because they depict many of the characteristics discussed earlier.

- Does not cannibalize existing services
- Can be offered at a premium price
- Provides differentiation
- Can provide synergy with basic service

Largely due to the current state of mobile communications evolution, many non-voice services can be considered to be value-added. However, the extent to which additional value-added services can be layered on top of mobile data services will determine the limit of their value.

For example, many non-voice services will have even greater value through personalization. Two of the most significant ways to personalize wireless services are through location enabling them and making them personal profile driven.

Mobile data services are utilized to obtain information, content, and to perform transactions. All of these activities are more meaningful if they are tailored to the individual. Location based services add value by way of putting the data into a location context for the user. Personal profiles further enhance the value through Personalization.

5.5.6. STK

STK, defined in GSM 11.14, defines an interface between the subscriber identity module (SIM card) and the cellular handset.

Consequently, applications loaded onto the SIM card can be run via the handset interface. Network operators need a computational platform to run data applications on telephone handsets. Although handsets do not provide a uniform platform; network operators have complete control over the SIM cards they supply to their customers. SIM cards are therefore the ideal vehicle to implement and distribute applications. The SIM card also provides built in security, which is useful for business applications.

Applications will be available for download over the air from network operators e.g. MPESA and ZAP, and users will be able to add and remove applications from the SIM card as needed. Sun has developed a Java card, which runs Java programs and uses the STK interface.

5.5.7. MMS

Designed for 3G (and beyond) networks, Multi-media Messaging Services (MMS) provides a technical solution of even richer media including text, sounds, images and video to MMS capable handsets. While EMS will perpetuate the rather proprietary architectures and interfaces, MMS will instead be the first mobile messaging service to utilize open Internet standards for messaging. Unlike EMS, which utilizes existing SMS capable terminals, MMS will require new mobile devices.

Messaging evolution toward MMS is tightly coupled with and dependent on supporting technologies such as Java, Mobile Station Application Execution Environment

(MexE), and Bluetooth. Other supporting technologies include location based service systems and advanced billing systems.

A key component of MMS is what is referred to as the MMS relay function, which is the conversion of a phone number to an IP address. Unlike with SMS, MMS bearers (such as GPRS) have an IP address associated with the phone. The MMS relay function allows the sender of a MMS message to address it to the phone number, while the system converts to an IP address for routing to the Multi-media Messaging Service Center (MMSC) and end-user. It is important to note that SMS is "not going away any time soon". In addition to the fact that MMS capable phones must be provided to the end-users, SMS will be used as an alert mechanism - to tell customers that they have some MMS content waiting for them at the MMSC.

Over time, MMS will also need to integrate with unified mes-

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saging and communications systems.

5.5.8. IM (Instant Messaging)

IM in the fixed environment is perhaps best exemplified by the likes of AOL, Microsoft, and Yahoo, which each have a large base of IM users that typically use IM on a fixed client, such as a laptop computer. Fixed IM includes a rich User Interface (UI) experience and high bandwidth.

- **Characteristics of MIM- Mobile Instant Messaging**

In a mobile environment, the user is constrained by bandwidth and the UI. However, the user has the advantage of mobility. MIM users will desire to message with other MIM users as well as fixed IM users on networks such as AOL, Microsoft and Yahoo.

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MIM allows users to address messages to other users (via various bearers) using an alias (or user name) and address book, allowing the sender to know when his/her "buddies" are available.

- **SMS versus MIM**

SMS is a bearer that can be used for MIM. By itself, SMS does not include alias capabilities nor does it allow for confirmation that the intended recipient is available. Being transaction based (rather than session based), plain SMS (without MIM) does not allow the send to have a high degree of confidence that the recipient will receive the message in real-time. MIM allows the community of users to register as being presence and/or available, allowing for more real-time text messaging and communications than would be possible with traditional mobile messaging.

- **Future of MIM**

Future MIM systems will be much more robust and pervasive, encompassing more valuable functionality and including interoperability between IM systems.

5.5.9. Unstructured Supplementary Services Data

Unstructured Supplementary Service Data (USSD) is a technology unique to GSM. It is a capability built into the GSM standard for support of transmitting information over the signaling channels of the GSM network. USSD provides session-based communication, enabling a variety of applications. USSD is defined within the GSM standard in the documents GSM 02.90 (USSD Stage 1) and GSM 03.90 (USSD Stage 2).

- **Key attributes:**
  - USSD is session oriented, unlike SMS, which is a store-and-forward, transaction-oriented technology.
  - Turnaround response times for interactive applications are shorter for USSD than SMS because of the session-based feature of USSD, and because it is NOT a store and forward service.
  - Users do not need to access any particular phone menu to access services with USSD- they can enter the Unstructured Supplementary Services Data (USSD) command direct from the initial mobile phone screen.
  - USSD commands are routed back to the home mobile network's Home Location Register (HLR), allowing for the

virtual home environment concept – the ability for services (based on USSD in this case) to work just as well and in exactly the same way when users are roaming.

- Unstructured Supplementary Services Data (USSD) works on all existing GSM mobile phones.
- Both SIM Application Toolkit and the Wireless Application Protocol support USSD.

- **USSD in Operation**

In operation, USSD is used to send text between the user and some application. USSD should be thought of as a trigger rather than an application itself. However, it enables other applications such as prepaid. In operation, it is not possible to bill for USSD directly, but instead bill for the application associated with the use of USSD such as circuit switch data, SMS, or prepaid.

5.5.10. Inter-carrier Messaging

Inter-carrier Messaging (ICM) - sometimes referred to as inter-operator or inter-network messaging - refers to the ability to transmit messages between mobile communications networks regardless of technologies involved (CDMA, GSM, iDen, PDC, or TDMA) and regardless of SMSC protocols deployed (CIMD, SMPP,UCP).

- **Why is ICM Important?**

ICM functionality enables messaging between networks in which otherwise would be constrained to only allow for intra-network messaging between subscribers belonging to the same network.

This capability is of particular importance for countries and regions of the world such as the Americas and Asia-Pacific, which contain mobile operators that employ various technologies. ICM enables mobile operators in those areas to enjoy the same benefits of inter-network messaging enjoyed throughout Europe on the homogeneous GSM networks.

- **How to Deploy ICM**

ICM may be deployed by the operator (within its on network) but the more effective solution is to utilize a third-party solution. This is due to economies of scale issues and the administrative impact of provisioning and administration of information, particularly in a country employing mobile number portability.

The basic function of ICM is to:-

- Determine the termination network based on evaluation of the destination address of the MDN (ANSI) or MSISDN (GSM), with ancillary GTT support provided to resolve destination addresses that reside within ported line ranges.
- Once the termination network is identified, the ICM function performs protocol conversion and reformatting of message as necessary.

- **Important Features of ICM**

Along with support for basic SMS feature such as segmentation and concatenation, the successful ICM function must support additional features such as message reply, message confirmation, spam controls, message activity monitoring and reporting capabilities.

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- **The Future of ICM**

The future is bright for ICM as wireless carriers in the Americas and Asia Pacific as SMS mobile origination capability is now ubiquitous across all digital networks. The ability for mobile operators to send alpha-numeric text messages amongst themselves will prove crucial in supporting geometric growth curves in messaging as experienced in Europe with GSM.

Carriers across international boundaries will likely inter-connect to leading service bureau providers of ICM services to enable truly global messaging.

SMS is just the beginning. As mobile networks evolve to support more Advanced Messaging capabilities, end-users will begin to engage in more than merely person-to-person text messaging. Exchange of multi-media information and interactive informa-

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tion and entertainment will become more of the rule than the exception.

5.6. Prepay Technology

There are many technologies involved in deploying mobile prepay (prepaid wireless service). They include:-

5.6.1. Point Solutions

Point solutions involve an adjunct piece of equipment that is used to provide call control, database functions, administrative functions and call processing. These are called ”point” solutions as all calls must traverse a single network element or ”point”. The advantage of a point solution is that they are typically less expensive to set up for an initial or limited deployment. However, they do not necessarily scale as well as other solutions when large numbers of prepay customers are added to the system. An example of a point solution would is a Service Node based solution, which are often deployed based on programmable switch technology.

5.6.2. ISUP based Solutions

ISUP based solutions prepay solutions typically utilize an innovative network kluge known as ISUP-loop back. With ISUP loop-back signaling, the SCP contains software logic to ”spoof” the mobile switch. The SCP tricks the mobile switch, making it appear to be signaling with another switch for call set-up via ISUP (as normal), when in fact it is performing database operations necessary to process prepay calls. ISUP based solutions of this type are considered to be a form of point solution as every call must go through the single point at the voice loop-back trunks.

5.6.3. Intelligent Network based Solutions

These falls into two categories: Proprietary IN and standards based IN solutions.

- **Proprietary solutions** are limited due to its proprietary (vendor specific) nature, roaming is problematic. Roaming can only occur between like switch types that have the same software logic and triggers armed. This includes implementations that utilize proprietary extensions of the INAP standard.
- **Standards based IN solutions** are enabled by the WIN

and CAMEL standards for ANSI-41 and GSM mobile networks respectively. These IN standards provide capabilities for TCAP based standard prepay deployment. They provide a major improvement over proprietary TCAP solutions by allowing roaming to take place wherever serving carriers support the necessary triggers and software logic. This means that, in order to provide service to prepay callers, the serving system must support WIN/CAMEL standards, even if the serving market does not support WIN/CAMEL based prepay for its own (home) customers.

5.6.4. Handset Solutions

These methods involve the placement of intelligence in a mobile terminal to determine the capabilities and account status of the prepay subscriber. Handset solutions fall into two broad categories based largely on mobile network standards: Smart Card based solutions, such as Subscriber Identity Module (SIM) solutions based on GSM, and proprietary handset based solutions.

5.6.5. Hybrid Solutions

A hybrid solution example would be exemplified by a SIM based solution for basic prepay call processing with a TCAP based IN complement for value-added services such as location sensitive billing. It is likely that many GSM operators will adopt this approach as they migrate to a full CAMEL based IN architecture over time.

5.6.6. Call Detail Record (CDR) based Solutions

CDR based solutions utilize a near real-time based process to track mobile usage by prepay customers. This solution has the advantage of low cost compared to other solutions. As CDRs are monitored after a call is placed, this solution can sometimes allow unbillable usage on the last call (prior to the system detecting a zero account balance and initiating a disconnection).

EXERCISE 1. ✍ Explain what is meant by value added services (VAS)

Example ✏. Explain the paging function of cellular networks

Solution: Paging is a simple and inexpensive form of mobile communication. Paging systems are designed to provide reliable communication to subscribers wherever they are. □

Exercise 2. ✍ List and explain two types of mobile communication technologies

Solutions to Exercises

Exercise 1.
VAS stands for Value Added Service and is a term that is used to refer to service options that are complimentary to a core service offering.

Exercise 1

Exercise 2.

Communication Satellites: Satellites consist of large transponders that listen to a particular radio frequency, amplify the signal, and then rebroadcast it at another frequency. They are inherently broadcast devices.

Cellular Radio Networks: Cellular networks are divided up into cells, each cell being serviced by one or more radio transceivers (transmitter/receiver). Communication in a cellular network is full duplex, where communication is attained by sending and receiving messages on two different frequencies. Exercise 2

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